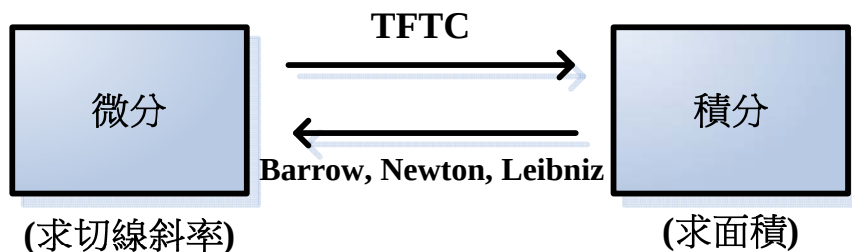


§5-3 The Fundamental Theorem of Calculus

Homework: 3,9,13,17,29,30,37,42,47,51,53,54,62,67



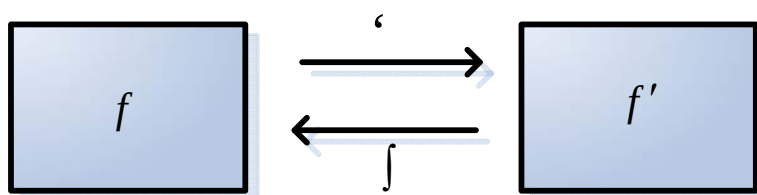
TFTC : Let f is constant on $[a, b]$

part I : If $g(x) = \int_a^x f(t) dt$,

$$\Rightarrow g'(x) = f(x).$$

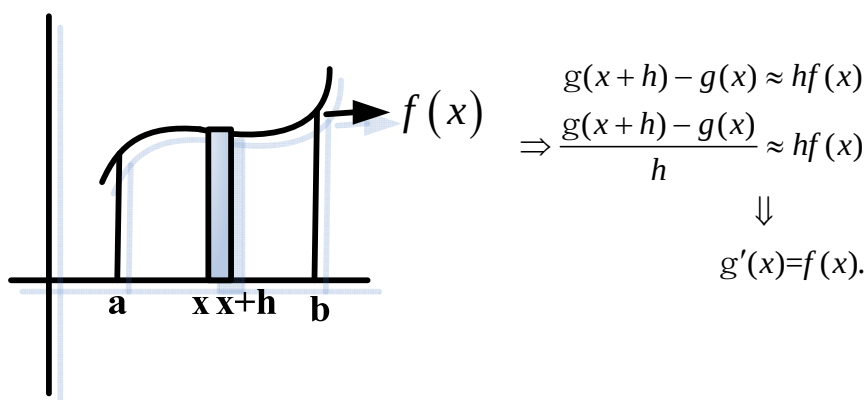
part II : $\int_a^b f(x) dx = F(x) \Big|_a^b = F(b) - F(a).$

where F is any antiderivative of f .



Geometric Intuition.

part I :



part II :

From part I $\Rightarrow g(x)$ is an antiderivative of f

$$\begin{aligned}\Rightarrow F(x) &= g(x) + c \\ \Rightarrow F(b) - F(a) &= (g(b) + c) - (g(a) + c) \\ &= g(b) - g(a) = \int_a^b f(t) dt.\end{aligned}$$

Example 1 : $\int_1^4 \frac{1}{\sqrt{x}} dx.$

Solution : $\int_1^4 x^{-\frac{1}{2}} dx = 2x^{\frac{1}{2}} \Big|_1^4 = 4 - 2 = 2.$

Find $\frac{dg}{du}$ in Example2- Example4

Example 2 : $g(u) = \int_3^u \frac{1}{x^2 + x} dx.$

Solution : $\frac{dg}{du} = \frac{1}{u^2 + u}.$

Example 3 : $g(u) = \int_u^3 \frac{1}{x^2 + x} dx = -\int_3^u \frac{1}{x^2 + x} dx.$

Solution : $\Rightarrow \frac{dg}{du} = -\frac{1}{u^2 + u}.$

Example 4 : $g(u) = \int_2^{\sqrt{u}} \frac{1}{x^2 + x} dx$
 $= F(\sqrt{u}) - F(2), \text{ where } F'(x) = f(x).$

Solution : $\Rightarrow \frac{dg}{du} = F'(\sqrt{u}) \frac{1}{2} u^{-\frac{1}{2}} = f(\sqrt{u}) \frac{1}{2} u^{-\frac{1}{2}}$
 $= \frac{1}{2\sqrt{u}} \left(\frac{1}{u + \sqrt{u}} \right).$

- $\frac{d}{dx} \int_a^{f(x)} g(t) dt = g(f(x)) f'(x).$
- $\frac{d}{dX} \int_{h(x)}^{f(x)} g(t) dt = g(f(x)) f'(x) - g(h(x)) h'(x).$

Example 5 : $g(x) = \int_{\tan x}^{x^2} \frac{1}{\sqrt{t + t^4}} dt$

Solution : $\Rightarrow g'(x) = \frac{2x}{\sqrt{2 + x^8}} - \frac{1}{\sqrt{2 + \tan^4 x} (1 + x^2)}.$

Example 6 : If $F(x) = \int_1^x f(t) dt$, $f(t) = \int_1^{t^2} \frac{\sqrt{1+u^4}}{u} du$,

find $F''(2)$.

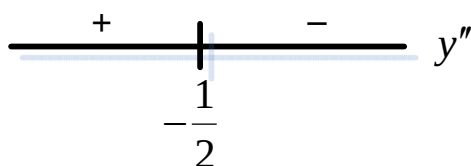
Solution : $F'(x) = f(x) \Rightarrow F''(x) = f'(x) = \frac{\sqrt{1+x^8}(2x)}{x^2}$

$$\Rightarrow F''(2) = \sqrt{1+2^8} = \sqrt{257}.$$

Example 7 : Find the interval on which the curve $y = f(x) = \int_0^x \frac{1}{1+t+t^2} dt$

is concave upward.

Solution : $y' = \frac{1}{1+x+x^2} \Rightarrow y'' = \frac{-(1+2x)}{(1+x+x^2)^2}$



$$\Rightarrow \left(-\infty, -\frac{1}{2}\right)$$

Example 8 : $\lim_{n \rightarrow \infty} \frac{1}{n} \left(\sqrt{\frac{1}{n}} + \sqrt{\frac{2}{n}} + \cdots + \sqrt{\frac{n}{n}} \right) = \int_0^1 \sqrt{x} dx$

$$= \frac{2}{3} x^{\frac{3}{2}} \Big|_0^1 = \frac{2}{3}.$$

Example 9 : Find a function f and a number a such that

$$6 + \int_a^x \frac{f(t)}{t^2} dt = 2\sqrt{x} \quad (1)$$

For all $x > 0$.

Solution : (i) $6 + \int_a^a \frac{f(t)}{t^2} dt = 2\sqrt{a}$ (將a代入(1))

$$\Rightarrow a = 9$$

$$(ii) \quad \frac{f(x)}{x^2} = x^{-\frac{1}{2}} \quad (\text{將(1)式兩邊微分})$$

$$\Rightarrow f(x) = x^{\frac{3}{2}}.$$